

Nirmalya Ghosh

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EXPERIENCE

Teradata

Seattle, WA - Remote

Senior Cloud Engineer

Mar 2026 - Present

- **Architected an end-to-end GCP Private Service Connect (PSC) automation framework** spanning ingress, egress, and interconnect tracks — automating HTTPS load balancer setup, **Cloud Armor** policy enforcement, TLS certificate issuance via Certificate Manager, and Cisco Valtix egress pipeline; reducing manual provisioning effort by **60%**
- **Led security hardening of Global DNS Service** following a formal Threat Model review — implemented **WAF** protections (SQL injection, XSS, command injection), **Route53** query logging to CloudWatch with automated alarms and SNS escalation, and **CloudTrail** auditing, delivering full audit visibility and reducing DNS-layer attack exposure
- **Designed and launched Azure Kubernetes network infrastructure** from the ground up — built **Terraform** provisioning for AKS subnets and transit connectivity, containerized the stack with production images in **JFrog Artifactory**, and developed **CI/CD pipelines** to automate image creation and publishing to Artifactory
- **Drove cross-team unblocking and knowledge transfer** — authored a **Confluence** onboarding wiki covering architecture, access provisioning, and SDLC workflows; resolved critical Azure/GCP blockers across multiple teams; and contributed to the **GCP POC** for PSC-based private connectivity for a major customer, enabling a successful proof of concept and paving the way for broader adoption of PSC in our product offerings

Teradata

Seattle, WA - Remote

Cloud Services Engineer

Jan 2024 - Mar 2026

- Developed and owned the **Kubernetes-level, multi-cloud Network Service**, delivering essential networking capabilities like **IP whitelisting**, GCP subnet creation, **Express Route**, and **VPN connectivity**.
- Built a plugin-based **credential rotation service** using the Template Method design pattern in Python, integrating with **HashiCorp Vault** to automate rotation for Valtix API keys and GCP service account keys — reducing credential downtime incidents by **75%** and strengthening security compliance.
- Designed and implemented network-related **Terraform** configurations to support and extend Network Service features, ensuring seamless integration and consistent infrastructure automation across multi-cloud platforms.
- Executed **Terraform state migration** from GitLab to AWS S3 backends across staging, pre-prod, and production environments, establishing durable remote state with DynamoDB locking.
- Configured and maintained **CI/CD pipelines** using **GitHub Actions** and **JFrog Artifactory**, enabling reliable artifact promotion and reproducible infrastructure deployments.

Teradata

Seattle, WA - Remote

Cloud Software Engineer Intern

June 2023 - December 2023

- Built a **Network Performance Evaluation Microservice** for SREs to detect and diagnose network issues, cutting resolution time by **40%** and improving operational efficiency by **30%**
- Developed **AWS Lambda functions** in Python with **Terraform** to automate VM provisioning across AWS and Azure, reducing manual provisioning time by **50%** and human error by **40%**
- Extended a **FastAPI** microservice to support AWS and Azure Capacity Reservation Groups (reducing infrastructure costs by **15%**) and expanded product availability to Azure *westus* and *westeurope*, growing the user base by **25%**

Teradata

Seattle, WA - Remote

Cloud Software Engineer Intern

June 2022 - September 2022

- Built a **FastAPI** service using **Boto3** to query AWS Service Quotas and compute utilization metrics, serving as a pre-check gate for customer cloud provisioning
- Developed automated **end-to-end API tests** with **Postman** integrated into **GitLab CI/CD**, validating functionality, performance, and reliability

EDUCATION

Master of Science: Computer Science (Software Engineering)

Seattle University

December 2023

Seattle, WA

- Award: Winner of ACM Club Project Competition, Winter '21, Werewolf Online Game

Bachelor of Arts: Geography

University of Washington

Sept 2016 - Mar 2020

Seattle, WA

- Award: Senior Year Dean's List
- Leadership: Community Manager - Plenum (Undergraduate Geography Research Journal)

SKILLS

- Python, FastAPI, REST API, Docker, Kubernetes, Terraform, HashiCorp Vault, Cloud Architecture & Orchestration, AWS (Lambda, S3, EC2, APIGateway, DynamoDB, Route53), Azure (VNet, Express Route, AKS), GCP (GKE, Cloud NAT, PSC), GitHub Actions, JFrog Artifactory, Shell Scripting, Javascript, ReactJS, NodeJS, C++, PostgreSQL, MySQL, Git, Linux